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| 28. | <p>When the expression <math>4x^3 - 12x^2 + 2x + h</math> is divided by <math>4x^2 - 1</math>, the remainder is <math>kx + 2</math>. Find the value of <math>h</math> and of <math>k</math>.</p> <p>Ans: <math>k = 3, h = 5</math></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 29. | <p>The term containing the highest power of <math>x</math> in the polynomial <math>f(x)</math> is <math>-2x^4</math>. Two of the roots of the equation <math>f(x) = 0</math> are <math>-\frac{1}{2}</math> and <math>1</math>. Given that <math>x^2 - 3x + 5</math> is a quadratic factor of <math>f(x)</math>, find</p> <ul style="list-style-type: none"> <li>(i) an expression for <math>f(x)</math> as a product of 3 factors,</li> <li>(ii) the remainder when <math>f(x)</math> is divided by <math>x</math>,</li> <li>(iii) the number of real roots of the equation <math>f(x) = 0</math>, justifying your answer.</li> </ul> <p>Ans: (i) <math>f(x) = (2x + 1)(1 - x)(x^2 - 3x + 5)</math> (ii) <math>f(0) = 5</math> (iii) 2 real roots</p> |
| 30. | <ul style="list-style-type: none"> <li>(i) Find the remainder when <math>3x^3 - x^2 + 12x</math> is divided by <math>3x - 1</math>.</li> <li>(ii) Hence, state the remainder when <math>3x^3 - x^2 + 12x + c</math> is divided by <math>3x - 1</math>.</li> <li>(iii) Factorise <math>3x^3 - x^2 + 12x - 4</math> completely, showing your workings clearly.</li> <li>(iv) Express <math>\frac{11x^2+9x+4}{3x^3-x^2+12x-4}</math> in partial fractions.</li> </ul> <p>Ans: (i) 4 (ii) <math>4 + c</math> (iii) <math>(3x - 1)(x^2 + 4)</math> (iv) <math>\frac{2}{(3x-1)} + \frac{3x+4}{(x^2+4)}</math></p>                                                                                                                                           |
| 31. | <p>The coefficient of <math>x^3</math> in a cubic polynomial <math>f(x)</math> is <math>k</math> and <math>f(x) = 0</math> has two distinct roots, 3 and <math>-2</math>. The remainder when <math>f(x)</math> is divided by <math>(x + 1)</math> is 4 times that of the remainder when <math>f(x)</math> is divided by <math>(x - 2)</math>. The graph of <math>y = f(x)</math> also passes through the point <math>(4, 12)</math>. Find <math>f(x)</math>.</p> <p>Ans: <math>f(x) = 2(x - 3)^2(x + 2)</math></p>                                                                                                                                                                                                                                    |
| 32. | <ul style="list-style-type: none"> <li>(i) Find the remainder when <math>3x^3 + 7x^2 + 5x + 2</math> is divided by <math>x + 1</math>.</li> <li>(ii) Factorise completely the cubic polynomial <math>3x^3 + 7x^2 + 5x + 1</math>.</li> <li>(iii) Express <math>\frac{-7x^2+7x+4}{3x^3+7x^2+5x+1}</math> as the sum of 3 partial fractions.</li> </ul> <p>Ans: (i) 1 (ii) <math>(x + 1)^2(3x + 1)</math> (iii)</p> $\frac{-7x^2+7x+4}{3x^3+7x^2+5x+1} = \frac{-3}{(x+1)} + \frac{5}{(x+1)^2} + \frac{2}{(3x+1)}$                                                                                                                                                                                                                                       |

33.

- (i) Find the remainder when  $8x^3 - 28x^2 + 42x - 8$  is divided by  $4x^3 - 16x^2 + 21x - 9$
- (ii) Factorise completely the cubic polynomial  $4x^3 - 16x^2 + 21x - 9$ ,
- (iii) Express  $\frac{8x^3 - 28x^2 + 42x - 8}{4x^3 - 16x^2 + 21x - 9}$  in partial fractions.

Ans: (i)  $4x^2 + 10$  (ii)  $(x - 1)(2x - 3)^2$  (iii)  
 $2 + \frac{14}{x-1} - \frac{26}{2x-3} + \frac{38}{(2x-3)^2}$